MACHINE LEARNING PROJECT · TECHNICAL REVIEW

Kickstarter Launch Check

Technical deep dive: data, model, evaluation and limits of a pre-launch funding estimate

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EXECUTIVE SUMMARY

Calibrated pre-launch chances that beat the goal rule

372,293

projects

2009 to early 2018, 36% funded

0.754

ROC-AUC (test)

Cross-validation 0.752; ranks 3 in 4 pairs right

0.619

F1 (test)

Goal rule: 0.524; always "not funded": 0

0.189

Brier score

Calibrated: predicted and actual within 1 point per bin

What drives success

- Category: Dance 62% vs Technology 20%, also within the same goal range
- Goal: 49% funded up to $1k, 6% over $100k (log-linear)
- Length: round numbers (30, 60 days) trail their neighbours

What to watch

- Time split (train <2016, test 2016+): ROC-AUC 0.757, rates drift
- Associations, not causes: no what-if advice
- No content, creator or marketing data

Held-out test set: 74,459 projects · Notebook: notebooks/01_kickstarter_eda_and_modeling.ipynb

PROBLEM FRAMING

One question, pre-launch inputs, probability metrics

1

Target

success = 1 if funded; 0 if failed, canceled or suspended. Live and undefined dropped.

2

Inputs

Only what a creator knows before launch: category, country, goal, length, launch month and weekday, title length.

3

Metrics

ROC-AUC, log loss, Brier and calibration for the probability. F1 for the yes/no label. Not accuracy: "always no" scores 64%.

Notebook: notebooks/01_kickstarter_eda_and_modeling.ipynb

DATA CLEANING

From 378,661 rows to 372,293 clean projects

378,661
raw projects
15 columns

→

372,300
known outcome
−6,361 live or undefined

→

372,293
valid launch date
−7 dated 1970

→

8
model features
pre-launch only

234 country codes

'N,0"' rows: country inferred from currency (EUR → "EU"). They were corrupted records.

24,135 name tags

"(Canceled)" and "(Suspended)" added after the outcome: stripped as leakage.

698 edge cases

Canceled projects that had reached the goal: kept as not funded (no money paid).

Kaggle dataset kemical/kickstarter-projects, file ks-projects-201801.csv (15 columns, no duplicate rows or IDs)

TARGET DEFINITION

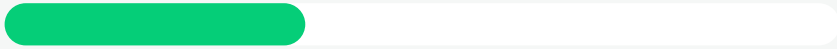
Funded, or not funded for any reason

state	projects	target
successful	133,956	**1 · funded**
failed	197,719	0 · not funded
canceled	38,779	0 · not funded
suspended	1,846	0 · not funded
live	2,799	dropped (unknown)
undefined	3,562	dropped (unknown)

Class balance after cleaning

36%

funded
133,956



Accuracy is misleading: "always not funded" scores 64%. No resampling and no class weights; the yes/no threshold handles the imbalance.

Notebook: notebooks/01_kickstarter_eda_and_modeling.ipynb

TARGET LEAKAGE

Remove the future data: everything known only after launch

Obvious: outcome columns

99.7%

of outcomes follow from "pledged ≥ goal"

- pledged, usd pledged, usd_pledged_real
- backers
- state (the raw label)

Hidden: tags in the name

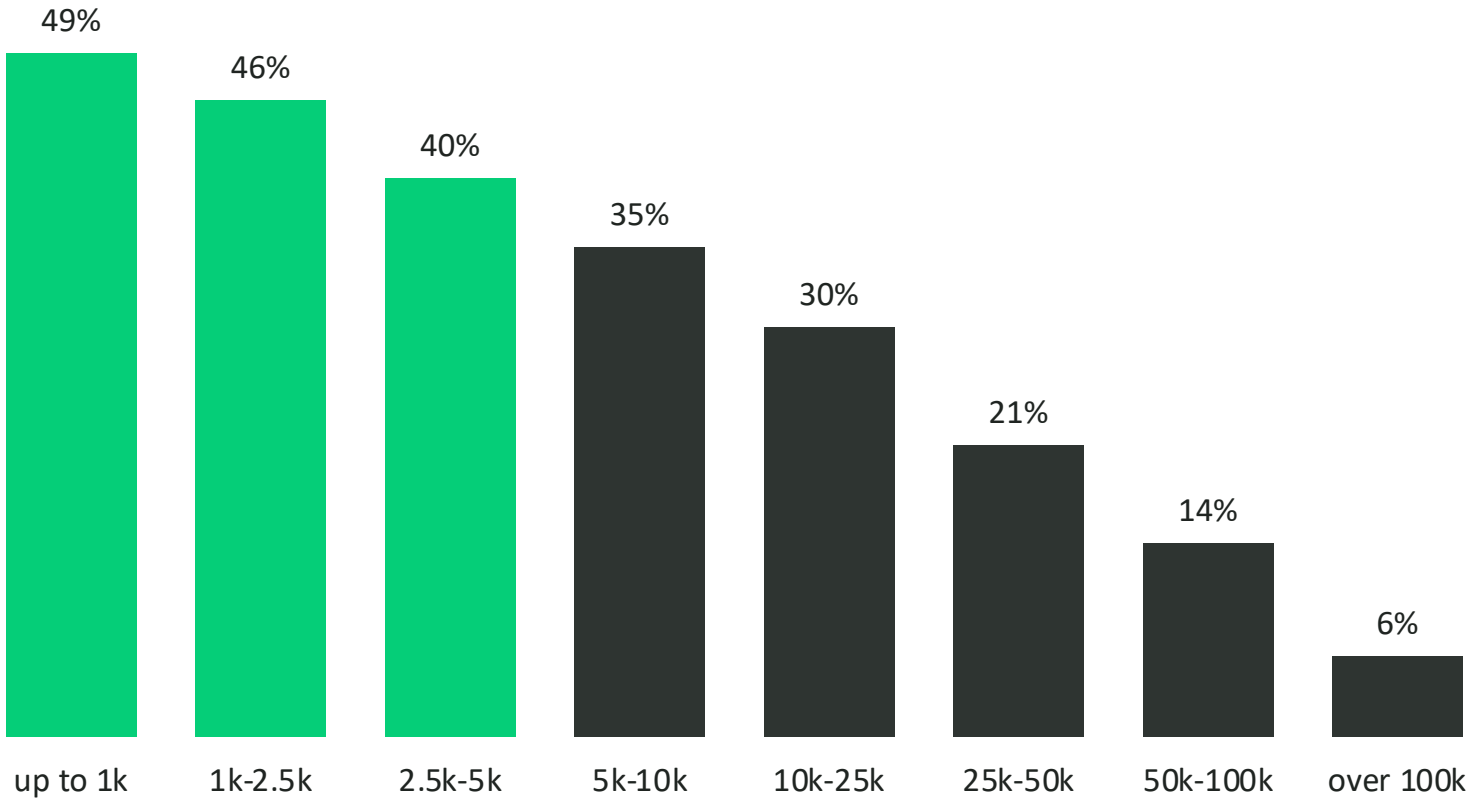
24,135

names ended in "(Canceled)" or "(Suspended)": 100% not funded

- Stripped with a case-insensitive regex before name features
- Residue accepted: ~120 renamed projects, 4 "(Cancelled)"
- Name features weigh little (importance ≤ 0.008)

99.7% on all 372,293 projects (canceled count as not funded); 99.997% among successful and failed only (alternative notebook, notebooks/reference)

Success falls by a similar step each time the goal doubles



Feature choice

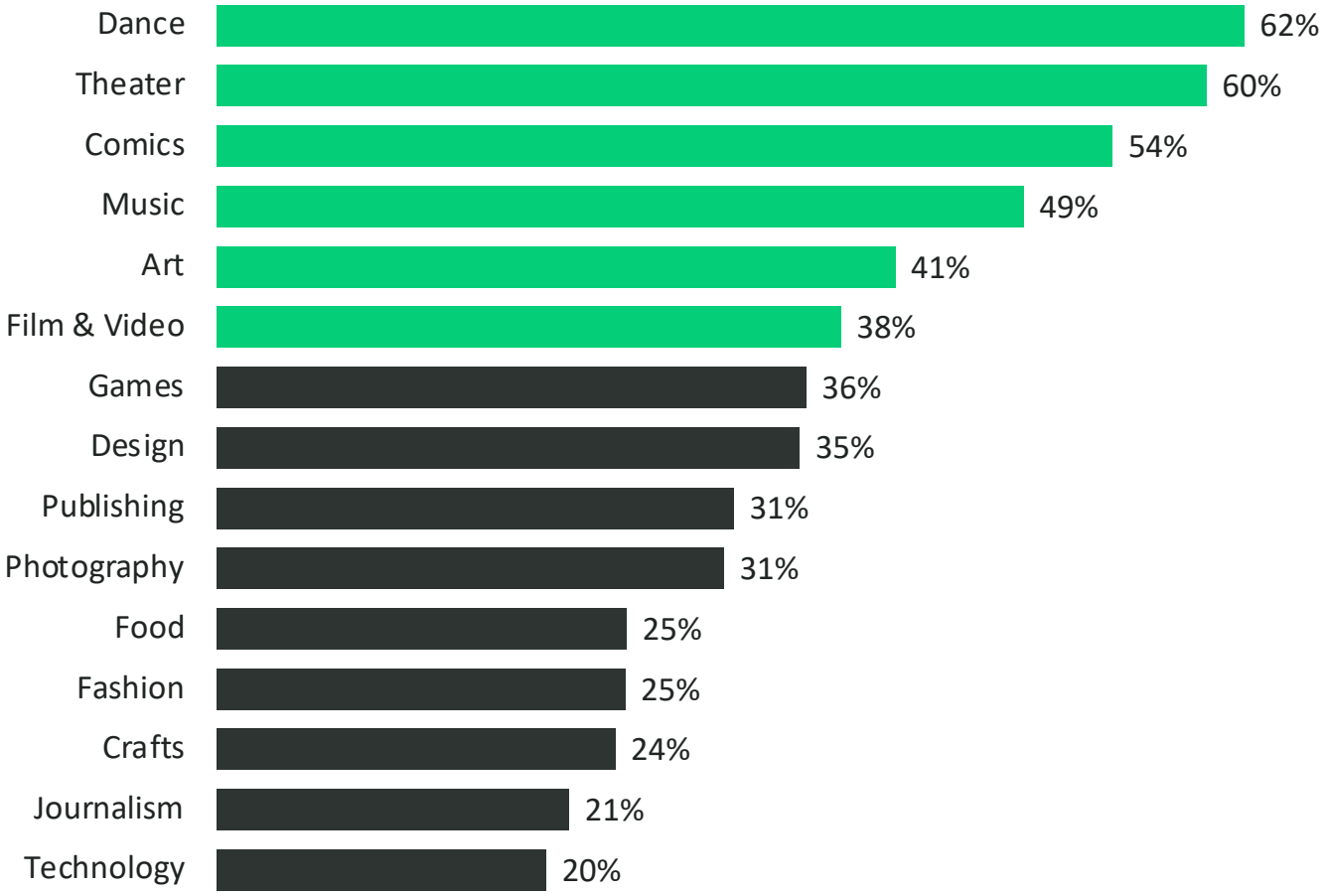
log_goal = log(1 + goal in USD). Goals run from $0.01 to $166M.

Robustness

1,143 goals ≤ $10 (63% funded). Without goals under $100, up to $1k is still 49.2%.

Success rate by goal in USD · all 372,293 projects · green: above the 36% average

Category matters, and not only through the goal



Goal $1k to $5k	funded
Dance	69%
Theater	68%
Technology	26%
Crafts	20%

Median goal: Technology $20,000, Dance $3,300, Crafts $2,328: low goal, still low success.

All 15 main categories · the model uses the 159 sub-categories · green: above the 36% average

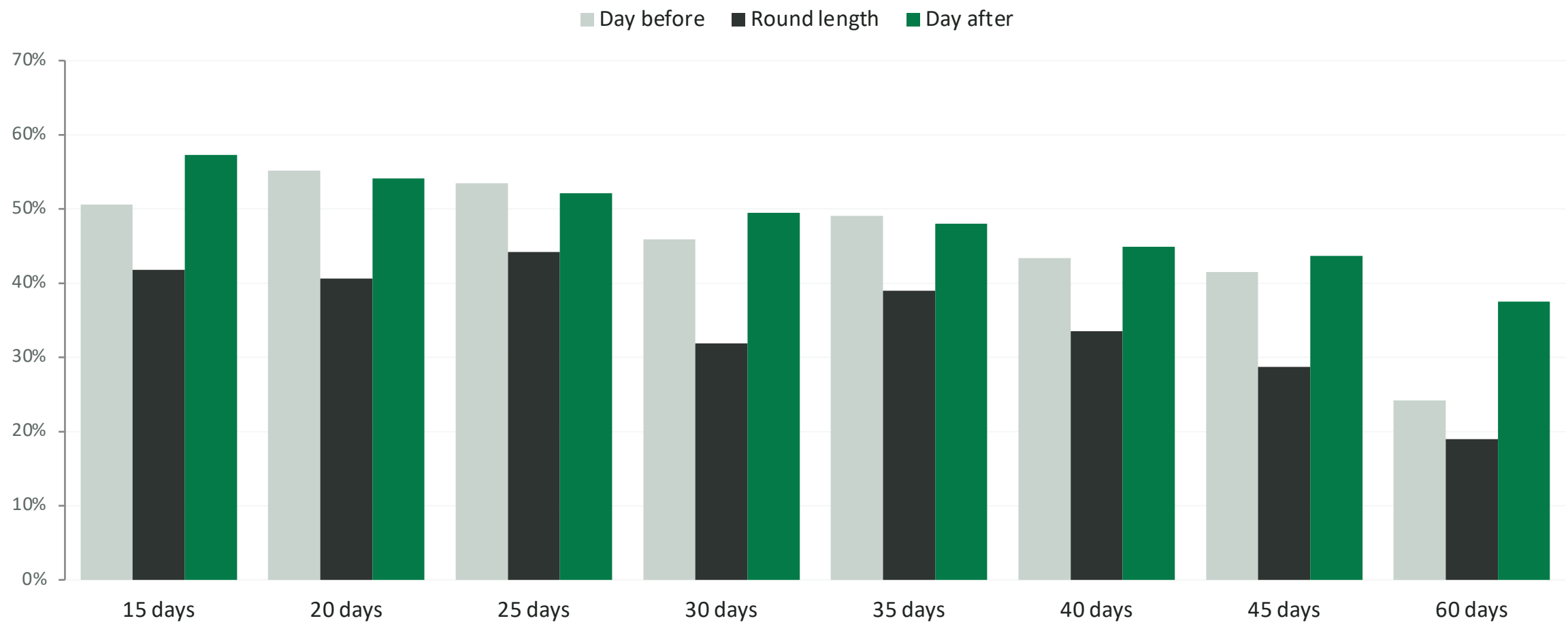
Compare lengths at the same goal size

Goal \ days	1-14	15-29	30	31-44	45-59	60	61-92
up to 1k	59%	59%	**43%**	57%	46%	**35%**	49%
1k-5k	46%	54%	**38%**	54%	39%	**26%**	40%
5k-25k	28%	42%	**29%**	43%	29%	**18%**	28%
over 25k	7%	22%	**14%**	26%	15%	**6%**	8%

Holds for every goal size: exactly 30 days is below both 15–29 and 31–44 days, and 60 days is the lowest column.

Success rate by goal range (rows) and campaign length in days (columns) · colour: coral below, green above the average

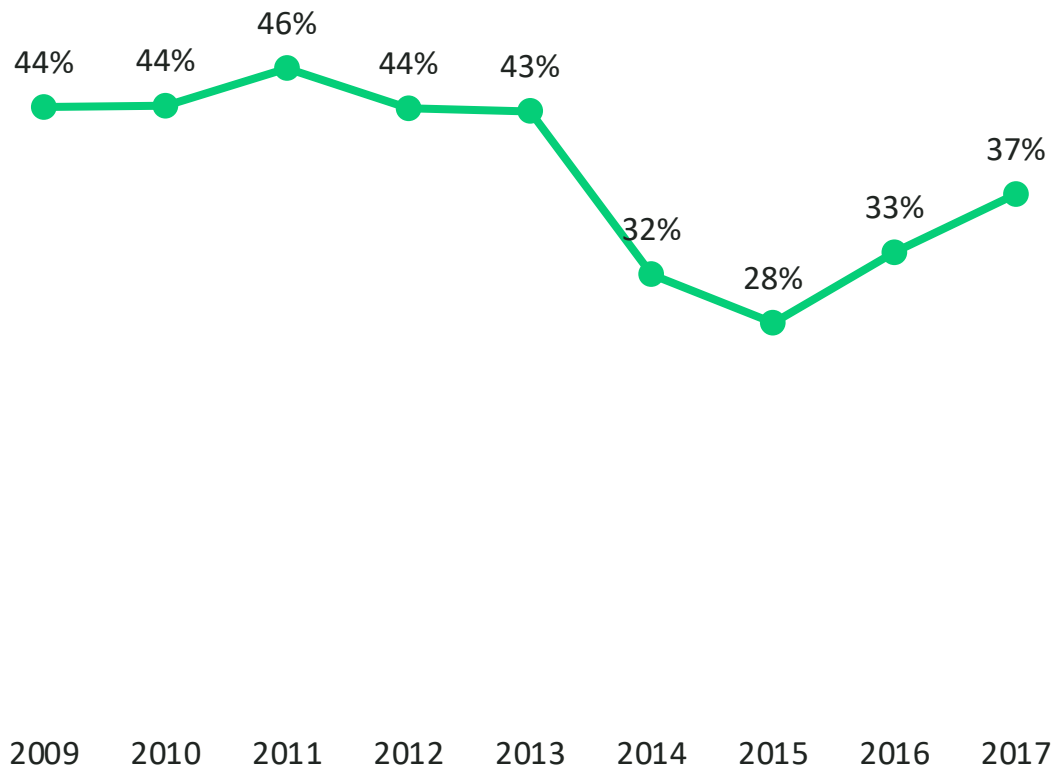
Every round length trails the day before and after



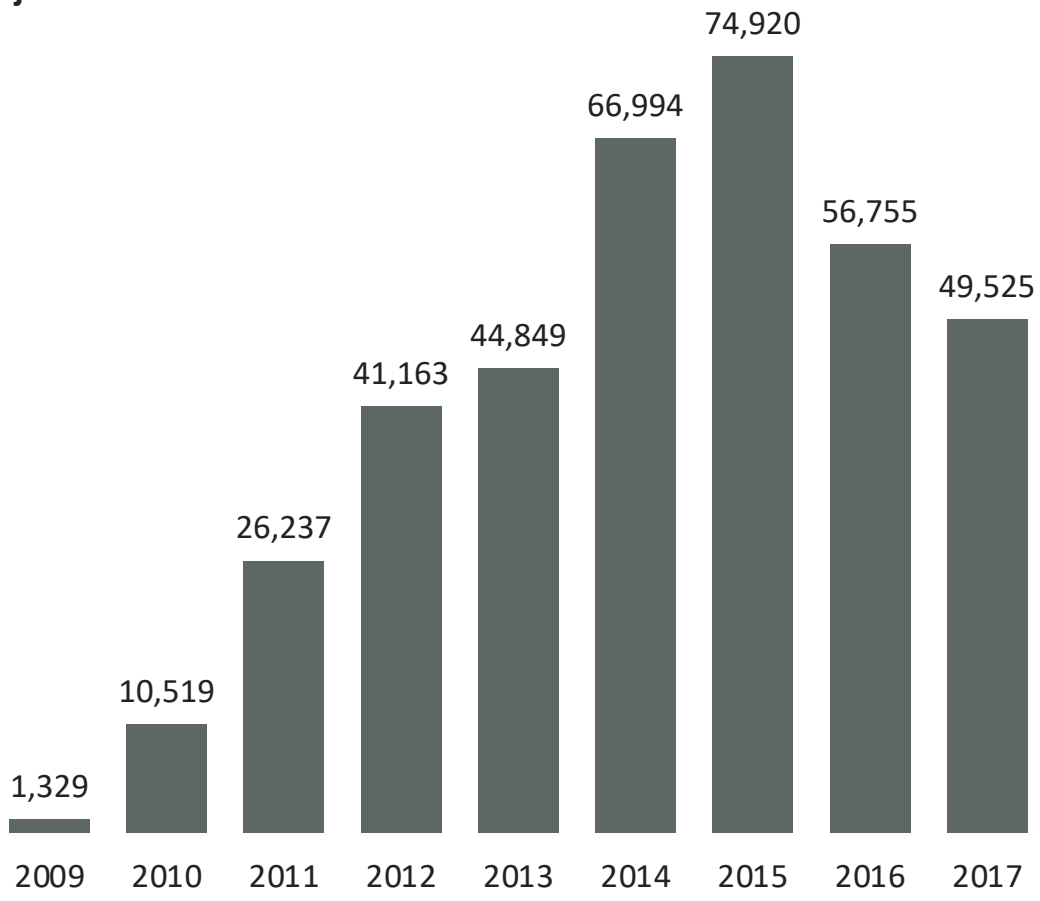
Success rate at each round length and its neighbours · same pattern inside each goal range (32 of 32 cases)

Success rates shifted over time

Success rate



Projects launched



2018 has only 2 projects (data collected in early January 2018) and is left out

FEATURES AND SPLIT

Eight pre-launch features, one untouched test set

feature	meaning
category	sub-category (159 values), native categorical
country	country code (23 values incl. EU)
log_goal	log(1 + goal in USD)
duration_days	deadline minus launch date, 1 to 92
launch_month	1 to 12
launch_weekday	0 = Monday to 6 = Sunday
name_length	title length in characters (tags removed)
name_words	title length in words

Train

297,834

Test (final exam)

74,459

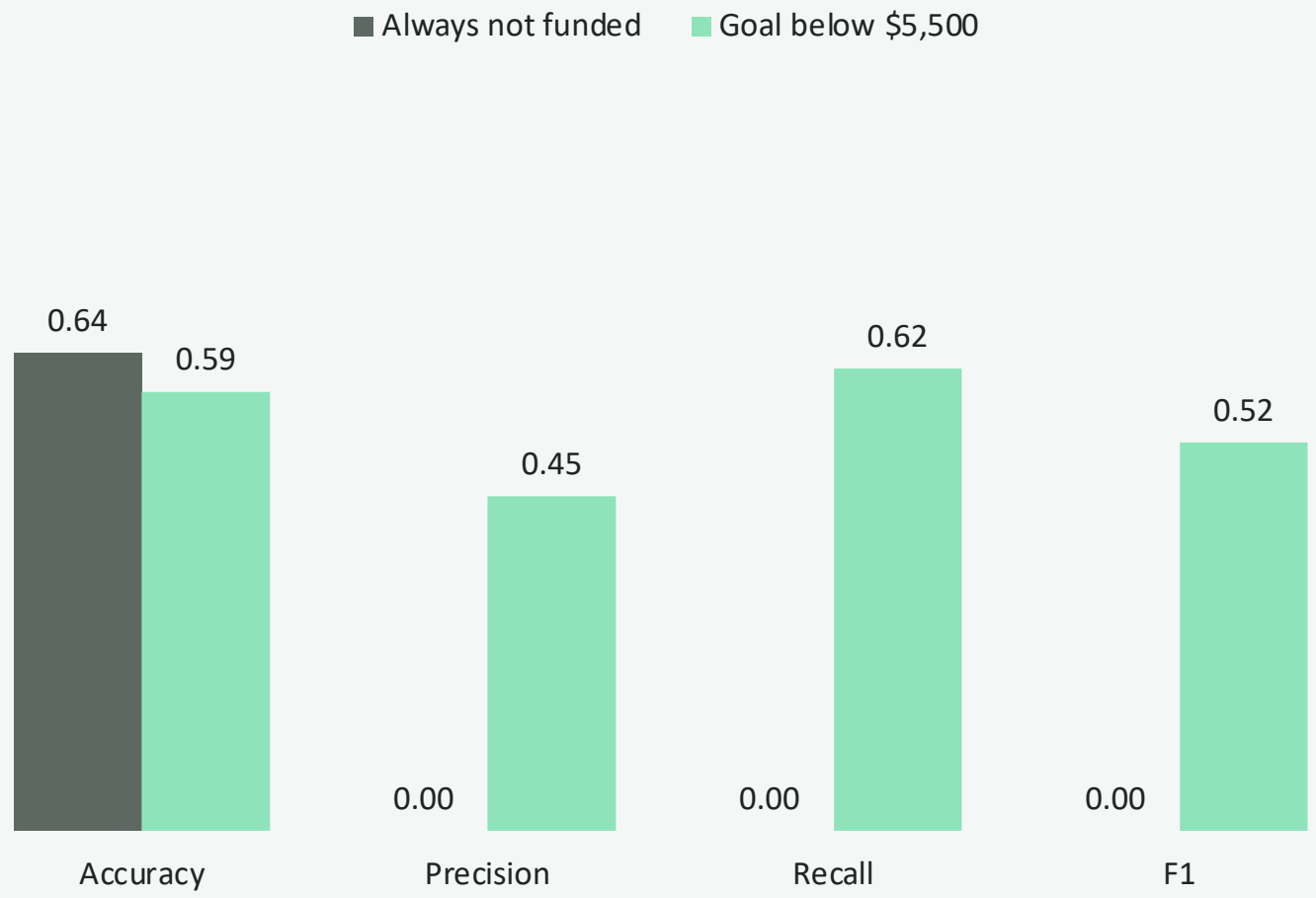
Left out on purpose

main_category (inside category), currency (= country), launch_year (future campaigns)

Stratified 80/20 split, random_state=42 · every encoder and model is fitted inside a Pipeline on training data only

BASELINE

Accuracy rewards doing nothing; F1 does not



0.520

Score to beat (F1)

± 0.002 across folds · test: 0.524

0.000

F1 of "always no"

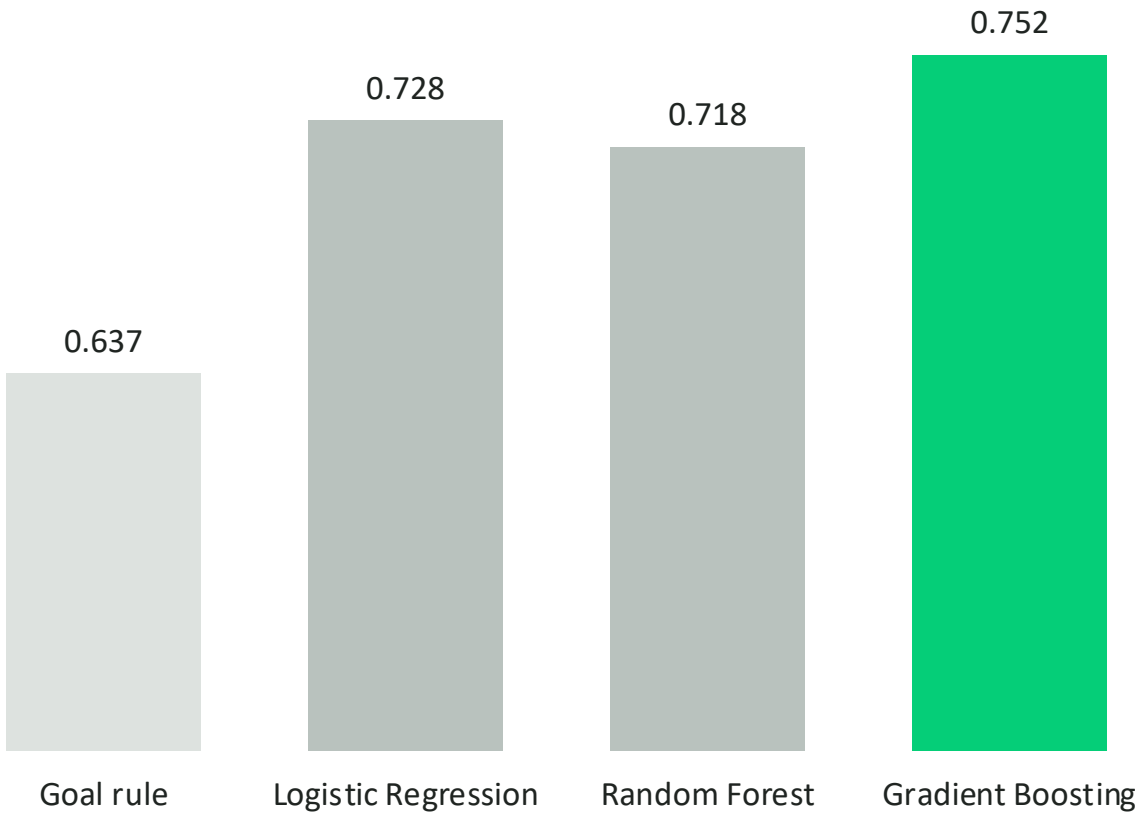
despite 64% accuracy

5-fold stratified CV on the training set · the goal median ($5,500) is learned inside each fold

MODEL COMPARISON

Gradient boosting performed best

ROC-AUC, 5-fold CV (axis starts at 0.50 = coin flip)



model	ROC-AUC	log loss	Brier
Logistic Regression	0.728 ± 0.001	0.575 ± 0.001	0.1971
Random Forest	0.718 ± 0.001	0.581 ± 0.001	0.1998
Gradient Boosting	0.752 ± 0.001	0.556 ± 0.001	0.1895

HistGradientBoosting with native categorical support; no class weights, so probabilities stay calibrated.

5-fold stratified CV on the full training set · mean ± std across folds (Brier std < 0.001) · Brier and log loss: lower is better

Tuning barely moves the needle; the threshold matters

hyperparameter	chosen
learning_rate	0.05
max_iter	400
max_leaf_nodes	31
min_samples_leaf	100
l2_regularization	1

0.5555

best CV log loss

defaults: 0.5558 (± 0.001)

0.299

F1-optimal threshold

CV F1 0.618

Early stopping is on by default, so max_iter is a ceiling (the final fit stopped at 231 trees). The threshold only sets the yes/no label; the probability is untouched.

RandomizedSearchCV (20 combinations, 5-fold, log loss) · TunedThresholdClassifierCV (5-fold, F1)

TEST SET RESULTS

Beats the goal rule, also at the same share

metric	Goal rule	Model (F1 threshold)	Model (same share)
Accuracy	0.591	0.641	0.671
Precision	0.451	0.501	0.530
Recall	0.625	0.811	0.736
F1	0.524	**0.619**	0.616
Share labelled	50%	58%	50%
ROC-AUC	–	**0.754**	same model
Brier	–	0.189	same model

CV and test agree

ROC-AUC 0.752 (CV) vs 0.754 (test). Brier 0.1895 vs 0.189.

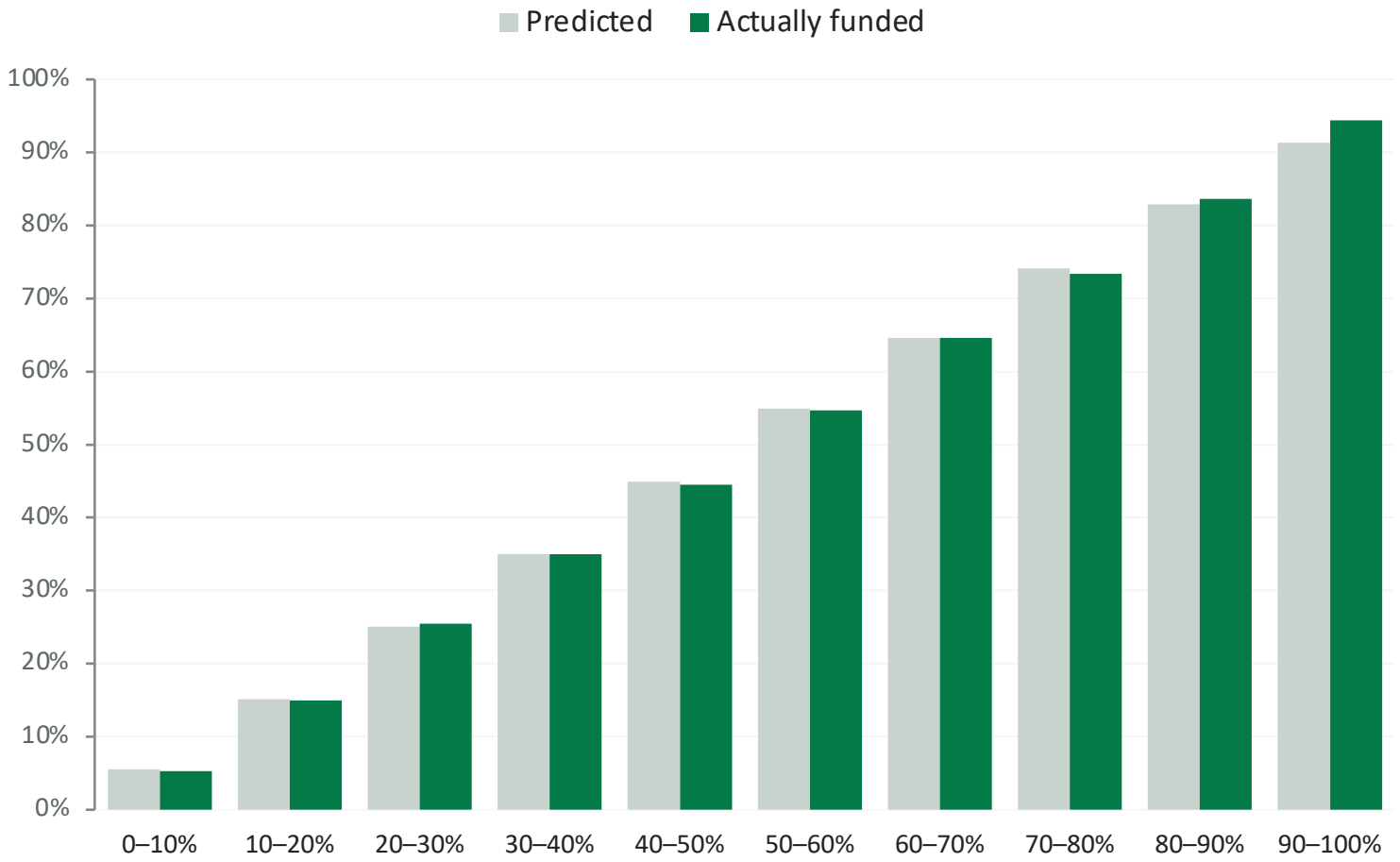
Warnings work

When it says "not funded" it is right 84% of the time (base rate 64%).

74,459 held-out projects · same share: the model's threshold set so it labels as many projects as the rule (50%)

CALIBRATION

When it says 40%, about 40% get funded



chance band	share	funded
below average	34%	**14%**
about average	39%	**37%**
above average	27%	**62%**

Bands are what Launch Check shows. Cut-offs at 25% and 50% are round numbers: always show the probability first.

Test set · 10 equal-width probability bins · the top bin holds only 18 projects

STABILITY OVER TIME

The ranking holds; the level drifts

0.757

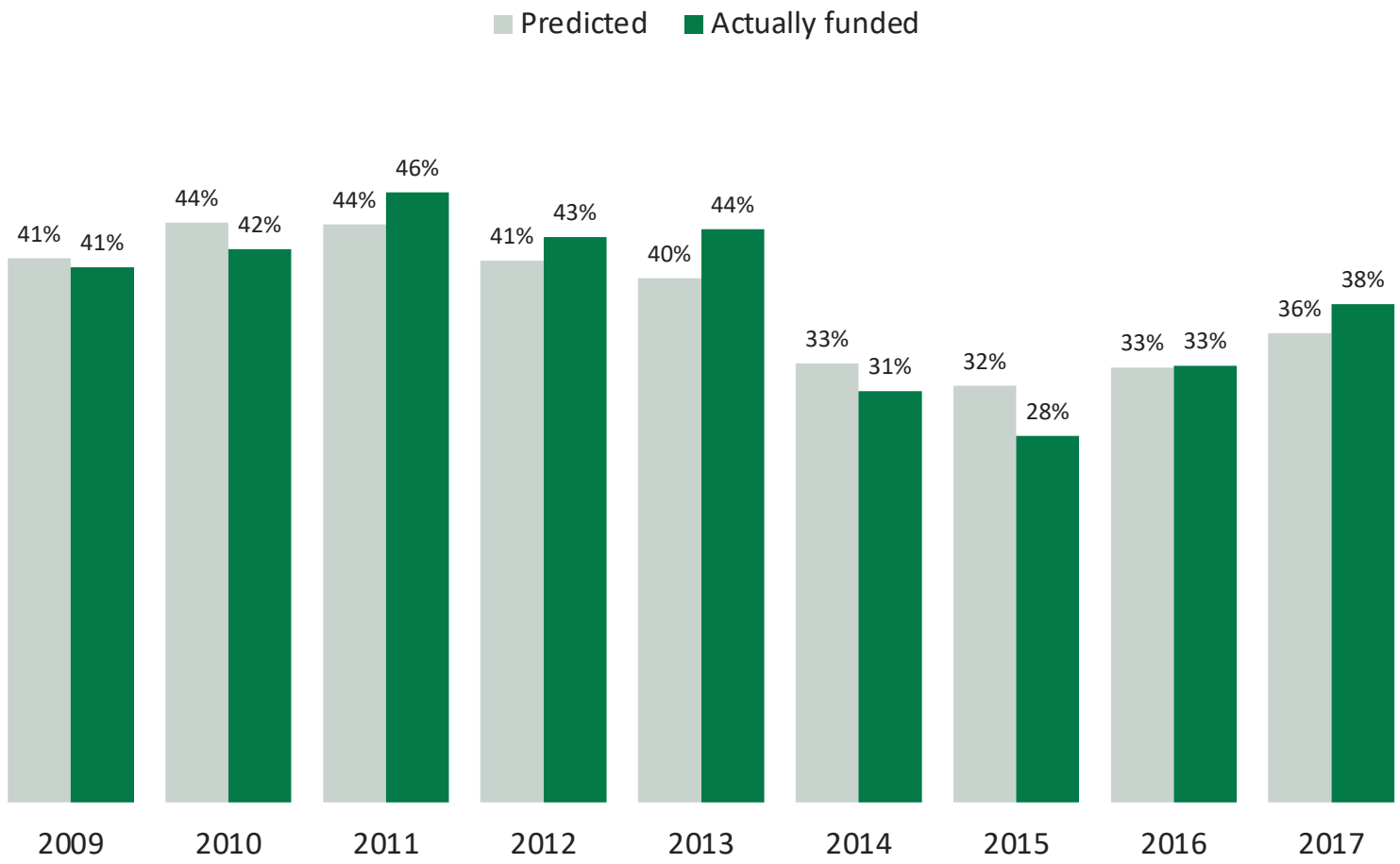
ROC-AUC, 2016+

random split: 0.754

33% vs 35%

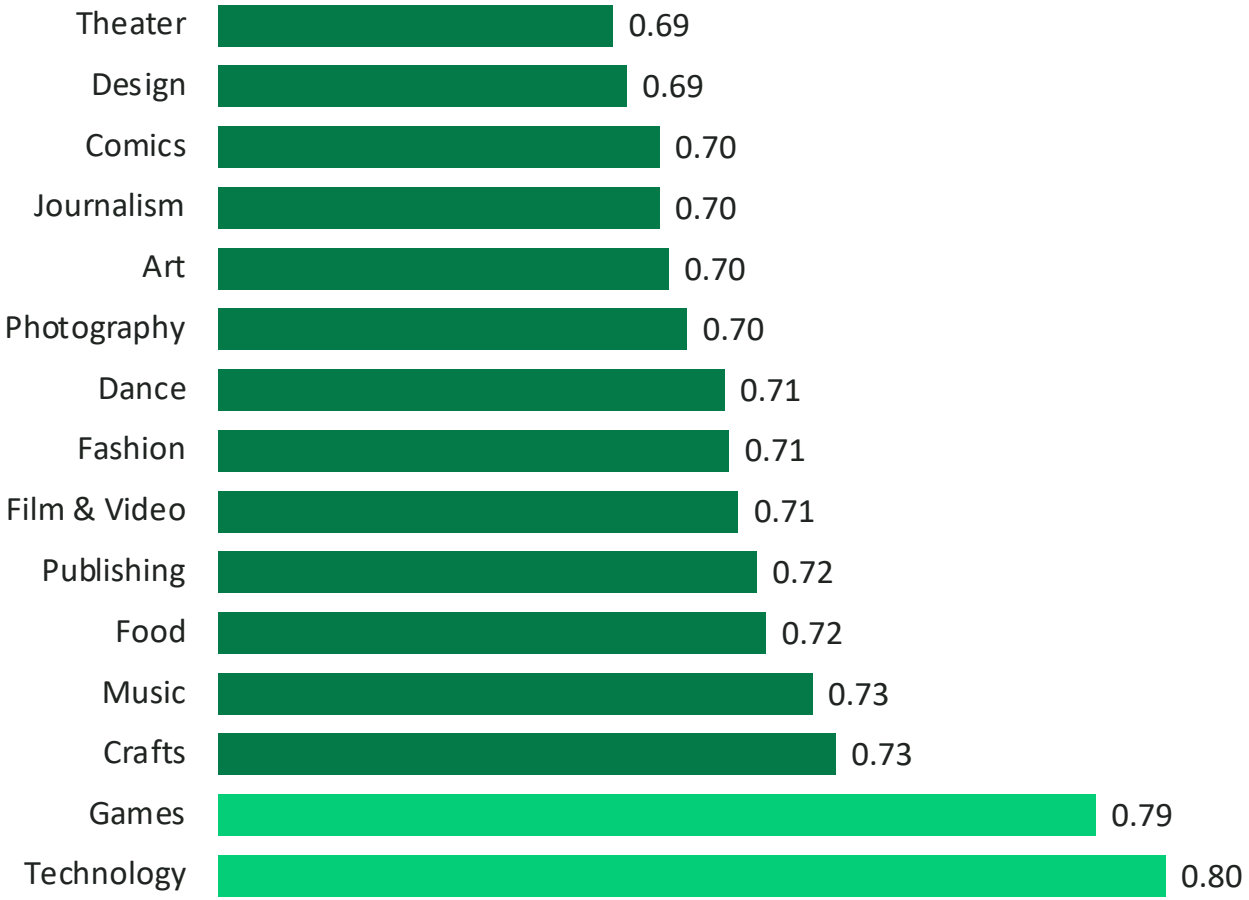
predicted vs actual

recall 0.749 (random split 0.811)



Left: trained on projects launched before 2016, tested on 2016 onwards · right: random-split test set by launch year

Probabilities hold in every segment; the label does not



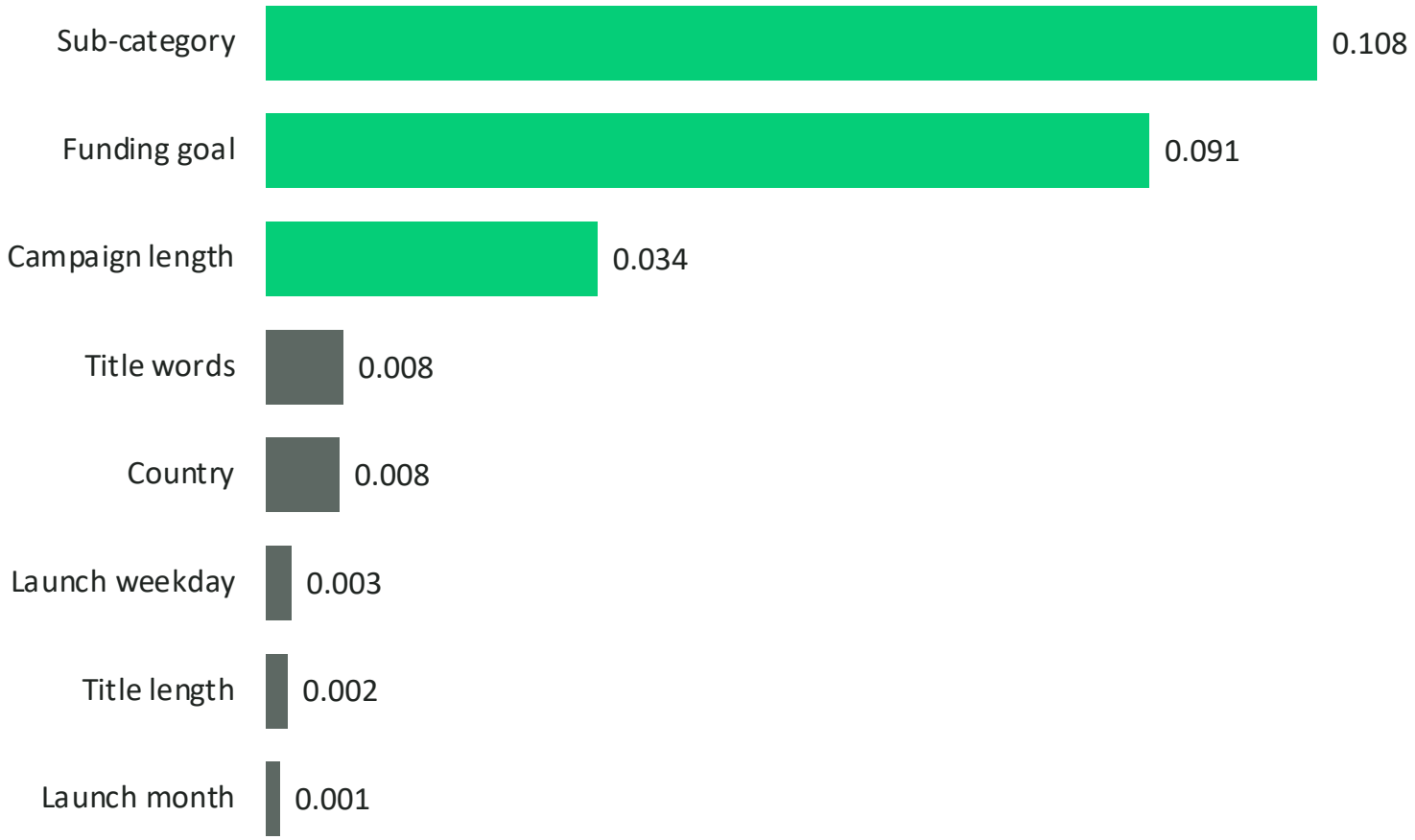
segment	funded	predicted	labelled
Theater	60%	60%	**91%**
Dance	63%	62%	**92%**
Goal ≤ $1k	50%	49%	**84%**
Goal > $100k	8%	8%	**2%**

Predicted matches funded everywhere; the yes/no label over-calls easy segments and under-calls big goals.

Test set · ROC-AUC within each main category (overall 0.754) · axis starts at 0.60

WHAT THE MODEL USES

Category and goal carry the prediction



Direction (partial dependence)

- Goal: probability falls steadily as the goal rises
- Length: low under ~10 days, highest at 3–6 weeks, sharp dips at round numbers
- Reliance, not cause

Permutation importance: drop in test ROC-AUC when one feature is shuffled (20,000 test projects, 5 repeats)

PRODUCT DECISION

Why Launch Check gives no what-if advice

scenario	predicted chance
As planned: 30 days, Tuesday	**51.1%**
29 days	**71.7%**
31 days	**62.1%**
Sunday launch	**38.3%**

Cross-check: similar past projects (Tabletop Games, $7.5k–15k, exactly 30 days): 1,375, of which 43% were funded.



Show the estimate only

- One-day changes swing the estimate by 10 to 20 points
- They reflect who picks those settings, not what the setting does
- If ever added: goal only, monotonic_cst, explicit warning

US Tabletop Games, $10,000 goal, launched on a Tuesday in March, 6-word title · saved model

TWO APPROACHES

Main analysis vs. the alternative notebook

	Main analysis (this deck)	Alternative notebook
Canceled / suspended	counted as not funded	dropped
Projects · funded	372,293 · 36%	331,675 · 40%
Launch year as feature	no (future campaigns)	yes, plus launch hour
Test ROC-AUC	0.754	0.765 (easier target)
Extra checks	calibration by year, time split, what-if audit	monotonic goal constraint
Idea to adopt	–	TF-IDF on title: CV ROC-AUC 0.765 → 0.782

Alternative: notebooks/reference/alt_approach_excluding_canceled.ipynb · its scores use a different target and rows, so they are not directly comparable

What to fix before real use

Limits

Associations, not causes

No what-if advice; round-number dips are behaviour

Blind spots

No video, rewards, creator network or marketing

Old data

Ends early 2018; per-year calibration off by up to ~4 points

Generous label

58% labelled promising vs 36% funded

Before real use

Refresh and recalibrate

Add recent projects; isotonic recalibration

Add text

TF-IDF on title and description (+0.017 CV AUC seen)

Add creator history

Past launches, backers, rewards

Goal-only what-if

monotonic_cst on log_goal, with a warning

Deploy and monitor

Launch Check as a web tool; track calibration drift

Recalibration: CalibratedClassifierCV(FrozenEstimator(model), method="isotonic") on the newest year, checked on a later slice

REPRODUCE IT

Everything runs locally in five commands

```
$ uv sync
# install Python 3.13 and pinned packages

$ uv run kickstarter-download
# fetch ks-projects-201801.csv into data/

$ uv run kickstarter-train
# clean, train, evaluate → models/ and reports/results.json

$ uv run kickstarter-check --category "Tabletop Games" --goal 10000 --days 30
# score a planned campaign

$ make slides
# rebuild both decks from reports/results.json
```

Requires uv (Python 3.13) · decks need Node.js 18+ · see README.md

Thank you for your attention.

Questions?